Jian Feng

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Education

University of Waterloo Sep 2023 - Aug 2027
Bachelor of Computing and Financial Management *Waterloo, ON, Canada*

Experience

Manulife Financial Corporation Toronto, ON, Canada
Software Engineer *May 2026 – Aug 2026*

- Built reusable pre-validation & skip-on-failure pattern across ServiceNow catalog task workflows (APIM, AKS, EMU onboarding), eliminating manual error handling and enabling safe async orchestration.
- Migrated SNOW orchestrator from basic auth to GitHub App token authentication via PyGithub, enabling secure mfc-ets team validation with audit-trail compliance across catalog automation.
- Enhanced GOCC orchestrator with Kubernetes command integration for live AKS cert-SAN verification, enabling the Foundry LLM agent to validate gitops PRs against cluster state and catch misconfigs before prod.

Treasury Board Secretariat Toronto, ON, Canada
Software Developer *Sep 2025 – Dec 2025*

- Shipped a .NET MVC module auto-normalizing job titles across a 5K-firm public dataset, unblocking analyst peer-group and governance reporting.
- Migrated 30 Selenium cases (5 suites) to Katalon on Azure DevOps CI/CD; demoed to execs and adopted by other QA teams to ease no-code onboarding.
- Refactored core modules of the enterprise data platform and tuned SQL, cutting 5K-firm-facing browser page load 30% for faster reviews.

Guangdong Tiptop Asset Management Co., Ltd. Guangdong, China
Analyst *Jan 2025 – Apr 2025*

- Built proprietary long screen on earnings quality, moat, and capex discipline; shortlisted BSE small-caps that fed the desk's weekly idea review.
- 3-statement + DCF on Audiowell (BSE: 920491, ¥3.5B mcap, 27% rev CAGR '22–'24); WACC × exit sensitivity framed an asymmetric base/bear IRR for the desk.
- Authored market-entry thesis on Hunan Smartcxi (commercial-kitchen EaaS): sized ¥50–200B TAM with labor-inflation + chain-penetration tailwinds.

Lansi Institute of Artificial Intelligence in Medicine Remote
Data Analyst *May 2024 – Dec 2024*

- Built genomic ETL pipelines for ~100GB multi-omics data; per-sample fan-out via Python `multiprocessing` cut runtime 89%.
- Ran XGBoost + SMR/MR² across a 68-study T2D meta-analysis as research assistant; narrowed the candidate drug-target shortlist used in the published paper.

Sanyang Technology Co., Ltd. Guangdong, China
Software Engineer *Jun 2024 – Nov 2024*

- Built Gantt visualizer of a flexible job-shop scheduling system, unblocking algorithm A/B runs on benchmark instances.
- Shipped Spring Boot CRUD APIs for cloud reporting/BI modules, unblocking front-end dashboards and downstream ETL consumers.

Skills

Languages:	Python, C/C++, Java, C#, Go, Bash, Typescript, Javascript, SQL, R
AI/ML:	PyTorch, TensorFlow, Hugging Face, LangChain, LangGraph, scikit-learn, Microsoft Agent Framework, OpenCV, pandas
Frameworks:	.Net, Spring Boot, React, Angular, Node.js, Flask, Rails, FastAPI
Databases:	PostgreSQL, MySQL, MongoDB, Redis
Cloud & DevOps:	AWS, GCP, Azure, Docker, K8s, Terraform, GitHub Actions, Linux, Jenkins
Tools:	Git, Jira, Confluence, Excel, PowerBI

Projects

CabStream ETL | Python

- Terraform-provisioned GCP stack (GCS + BigQuery + Airflow); whole pipeline tears down and rebuilds from one `apply`.
- dbt three-layer model (staging → dims/facts → marts) with `schema.yml` freshness/uniqueness tests; Airflow DAG logic unit-tested in CI.
- External-table-then-CTAS pattern keeps raw parquet cheap on GCS while marts stay query-optimised in BigQuery.

ArmyOfSafeguards

- Four DeBERTa safeguard heads (toxicity, sexual, jailbreak, factuality) run in parallel via `expert_runner`, hot-swappable via env vars.
- Learned logistic meta-aggregator fuses per-expert probabilities into a calibrated verdict, beating threshold baselines on held-out HF.
- Training pipeline turns HF labels into expert-Q features and meta JSONL, so adding a new expert needs no aggregator changes.

GhostPilot

- Invisible Windows overlay streams WASAPI loopback audio through partial/final Azure ASR into a text LLM.
- Alt+P vision path: Qt area capture → JPEG → vision LLM, result folded back into the text-LLM prompt.
- Local RAG over `knowledge/` injects user-curated context so answers stay grounded across long sessions.

Achievements

- Scaled UX Research Hub 32× (10→318 members) as Director; negotiated Google UXR keynote at $0 cost and brokered TikTok chapter cross-promotion for sustained recruitment growth.
- Velocity Co-op Problem Award (top submission among co-op cohort); 3rd Place, SAF Investment Research Challenge; Finalist, SAF Strategy Case Competition, Moonshot Hackathon Winner

Coursework

Google Data Analytics, Stanford Machine Learning, UCI Database Design & Operational BI, UPenn Finance & Quantitative Modeling, UCSC Bayesian Statistics